

Cem Ozcelik

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PROFESSIONAL SUMMARY

Data Scientist with an Industrial Engineering background and nearly 5 years of end-to-end ML experience spanning anomaly detection, financial risk modeling, payment intelligence, and MLOps deployment. The Industrial Engineering foundation - operations research, MILP, cost-function optimization - drives every model decision: I ask what it costs when the system is wrong in each direction before touching architecture. Proof: 15% OEE gain via unsupervised anomaly detection on CNC sensor data, 8-hour weekly overtime reduction through ML-driven supply chain forecasting, and a full credit risk MLOps pipeline (XGBoost, Docker, Django) built around the same cost-matrix logic that governs payment fraud. Proficient in Python (scikit-learn, TensorFlow, PyTorch, h2o.ai, XGBoost), advanced SQL, and containerized model deployment via Django/FastAPI.

CORE COMPETENCIES

Skills: Anomaly Detection, Financial Risk Modeling, Payment Intelligence, Statistical Analysis & Data Mining, A/B Testing & Experiment Design, ML-Ops Infrastructure, Business Insights Communication, Explainable AI

Tools: Python, scikit-learn, TensorFlow, PyTorch, h2o.ai, XGBoost, SQL (Advanced), Docker, FastAPI, Django, Flask, PySpark, Databricks, AWS

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 - Present

Senior Database Analyst | Maltepe, Istanbul

- Scaled a custom Django web application into a **multi-tenant ML deployment platform** with versioned REST APIs, removing all ad-hoc script deployments and enabling new models to reach production in days.
- Directed predictive supply chain optimization using Python-based forecasting algorithms, sustaining an **8-hour weekly reduction in operational overtime** and providing business teams with actionable ML-driven insights.
- Translated SKU-level Association Rule Mining outputs into strategic cross-sell actions, driving continuous revenue growth.
- Tech:** Python, Django, REST APIs, ML pipelines, Association Rule Mining, PostgreSQL, Docker, h2o.ai

Anadolu Pet Akvaryum & Effeffe

Sept 2024 - Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Architected a full-stack internal web application from scratch (Python/Django, SOLID principles) - establishing the deployment foundation later scaled into the ML platform.
- Engineered predictive supply chain optimization models in Python and PL/SQL, **reducing stockouts and improving inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, defining cost-to-profit ratios that delivered a **3% improvement in logistics efficiency**.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 - Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Built a **Predictive Maintenance & Anomaly Detection framework** using unsupervised clustering on machine sensor time-series - the same statistical approach applies directly to payment transaction anomaly detection - delivering a **15% weekly increase in critical machine availability**.
- Bridged raw production sensor data and operational strategy, **improving Overall Equipment Effectiveness (OEE) by 15%**.
- Engineered ETL pipelines in MSSQL and Python to ingest high-volume manufacturing sensor data, replacing manual reporting workflows.
- Processed complex time-series data for cycle time optimization, building a foundation for sequential modeling (RNNs).
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical analysis to workforce datasets, uncovering patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

End-to-End Credit Risk Prediction MLOps & Financial Risk

XGBoost classifier on Lending Club dataset under heavy class imbalance (<15% defaults), explicitly optimizing for asymmetric cost (FP = customer friction, FN = financial loss) - the same cost structure governing payment fraud detection. Full MLOps pipeline: raw data ingestion → feature engineering → containerized production app. Interactive Django EDA platform for non-technical stakeholders. **63% F1-Score** on temporal holdout (no leakage). Architecture transfers directly to payment anomaly detection.

Python, XGBoost, CatBoost, Django, Docker, scikit-learn | github.com/ccemozclk/Credit-Risk

Retail Demand Forecasting End-to-End MLOps Pipeline

~**12.83% RMSPE** on 6-week daily sales across 1,115 Rossmann stores with time-based validation. Object-oriented training pipeline (DataIngestion → DataTransformation → ModelTrainer). FastAPI REST inference + Streamlit dashboard with dynamic confidence intervals translating ML metrics to business terms. Surfaced actionable conversion insight: B-type stores yield lower promotion ROI vs A/C - enabled marketing budget reallocation.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossmann-Demand-Forecasting

AI-Powered Medical Diagnostics Explainable AI & Cost-Function Optimization

97% Recall on pneumonia detection from chest X-rays - optimized for asymmetric error cost (FN >> FP in clinical context). Fine-tuned VGG16 at **93% test accuracy** vs 76% baseline CNN. Implemented Grad-CAM for visual explainability; identified and corrected a dataset bias invisible without heatmap analysis. Demonstrates the same cost-function first principles applied to every high-stakes ML problem.

Python, TensorFlow-Keras, VGG16, ResNet50, OpenCV, Grad-CAM | github.com/ccemozclk/Deep-Learning-Project-CNN-VGG16-ResNet50

EDUCATION

B.Sc. Industrial Engineering — Kırıkkale University Sept 2016 - Feb 2021

Rigorous foundation in statistical analysis, linear algebra, and optimization. Designed MILP models with IBM ILOG CPLEX. Applied AHP and TOPSIS to multi-criteria decision problems - direct prerequisite for loss-function design and model tradeoff thinking in ML.

CERTIFICATIONS

Associate Data Scientist Career Track — [DataCamp](#) Jan - May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — [Miuul](#) Mar - Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash

ML & Statistical: scikit-learn, TensorFlow, PyTorch, XGBoost, h2o.ai, LightGBM, CatBoost, CNNs, RNNs, Transfer Learning, Grad-CAM, A/B Testing, Statistical Analysis, Data Mining

MLOps & Deployment: Docker, FastAPI, Django, Flask, REST APIs, Git, Linux **Big Data & Cloud:** PySpark, Databricks, AWS (EC2, S3)

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines

Advanced: Cost-Function Optimization, MILP, AHP/TOPSIS, OOP, System Architecture **Languages:** Turkish (Native), English (Fluent)